

- 1 (a) The enzyme alanine transaminase (ALT) is found in the liver. The function of ALT is to convert the amino acid α -ketoglutarate into another amino acid, glutamate.

Suggest why the liver may need to convert one amino acid into another.

.....
..... [1]

- (b) Tumours can form in the liver.

Explain how a liver tumour develops.

.....
.....
..... [2]

- (c) ALT can leak into the blood from liver tumour cells.

An increase in the concentration of ALT in the blood causes a decrease in the water potential of the blood.

State precisely the name **and** location of the cells where a change in the water potential of the blood would be detected.

.....
..... [1]

- (d) Describe the homeostatic role of ADH when the water potential of the blood **decreases**.

.....
.....
.....
.....
.....
.....
.....
.....
.....
..... [5]